Customer Churn Prediction & Lifetime Value Analysis

An End-to-End Machine Learning Project Report

This report documents a complete machine learning workflow applied to an e-commerce customer dataset, covering data preparation, regression and classification modeling, model evaluation, overfitting analysis, feature importance interpretation, and deployment as an interactive dashboard.

1. Introduction & Objective

Customer churn and lifetime value are two of the most important metrics for any subscription-based or e-commerce business. Predicting which customers are likely to churn allows a company to intervene early with retention offers, while estimating lifetime value helps prioritize marketing spend toward the most valuable customers.

The objective of this project is to build, compare, and interpret multiple machine learning models that address both problems using the same underlying customer dataset, and to deliver the final models through an interactive dashboard that a non-technical user can operate.

2. Dataset Description

The dataset used is **ecommerce_customer_churn_dataset.csv**, containing 25 columns describing customer demographics, engagement behavior, purchase history, and account status. The two target variables used in this project are:

- **Churned** (binary) — used as the classification target
- **Lifetime_Value** (continuous) — used as the regression target

2.1 Feature Engineering

Four additional features were engineered from the raw columns to better capture customer behavior patterns:

Engineered Feature	Formula
Purchase_Frequency	Total_Purchases / (Membership_Years + 1)
Engagement_Score	Login_Frequency + Pages_Per_Session + Social_Media_Engagement_Score
Customer_Activity_Score	Login_Frequency + Email_Open_Rate + Mobile_App_Usage
Risk_Score	Cart_Abandonment_Rate + Returns_Rate + Days_Since_Last_Purchase

3. Methodology

The workflow followed standard supervised learning practice: data cleaning, train/test splitting (80/20), preprocessing via a scikit-learn ColumnTransformer (mean/most-frequent imputation, standard scaling for numeric features, one-hot encoding for categorical features), model training inside a Pipeline, and evaluation on the held-out test set.

3.1 Regression Task — Predicting Lifetime Value

Two regularized linear regression models were trained: **Ridge Regression** (L2 penalty) and **Lasso Regression** (L1 penalty). Lasso's ability to shrink coefficients to exactly zero makes it useful for identifying which features contribute little to predicting customer value.

3.2 Classification Task — Predicting Churn

Two classifiers were trained on the same preprocessed features: **Logistic Regression**, a linear probabilistic baseline, and a **Decision Tree Classifier**, capable of capturing non-linear feature interactions.

4. Model Comparison

Classification models were compared using Accuracy, Precision, Recall, F1-score, and ROC-AUC on the test set. Regression models were compared using RMSE and R^2.

Note: replace the placeholder values below with the actual numbers printed by your notebook's comparison table before finalizing this report.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	—	—	—	—	—
Decision Tree	—	—	—	—	—

Model	RMSE	R^2
Ridge	—	—
Lasso	—	—

4.1 Discussion

Logistic Regression

- Interpretable coefficients — each feature's effect on churn probability is directly readable.
- Simple and computationally efficient, making it a strong baseline.
- Produces calibrated probabilistic output, useful for risk scoring and threshold tuning.
- Limitation: assumes a linear decision boundary and struggles with complex feature interactions.

Decision Tree

- Naturally captures non-linear relationships and feature interactions.
- Requires no feature scaling and produces human-readable decision rules.
- Limitation: prone to overfitting without depth control, and sensitive to small changes in the training data.

5. Overfitting Analysis

To study how model complexity affects generalization, Decision Trees were trained with max_depth ranging from 1 to 20, and training vs. validation accuracy were plotted at each depth (see accompanying notebook figure).

At low depths, both training and validation accuracy are low and close together — this is **underfitting**, where the model is too simple to capture the underlying pattern. As depth increases, training accuracy climbs steadily since the tree begins memorizing the training data, while validation accuracy peaks and then plateaus or declines. The depth at which the validation curve peaks marks the point beyond which the model is **overfitting** — learning noise specific to the training set rather than patterns that generalize to new customers.

6. Feature Importance Comparison

Feature importance was extracted from the Decision Tree (Gini importance) and compared against the magnitude of Logistic Regression coefficients. The top-5 features from each model were compared for overlap.

Based on the dashboard's live prediction explanation, the most influential features for churn risk consistently include **Lifetime_Value**, **Cart_Abandonment_Rate**, **Customer_Service_Calls**, **Discount_Usage_Rate**, and **Customer_Activity_Score**.

A high overlap between the two models' top features suggests both algorithms are detecting the same underlying signal, despite using very different mathematical approaches (linear coefficients vs. tree splits). Any divergence highlights non-linear interactions that the Decision Tree captures but the linear model cannot represent directly.

7. Interactive Dashboard

The final models were deployed in a Streamlit web application that allows a user to enter a customer's profile (demographics, engagement metrics, and purchase behavior) and receive:

- A churn probability percentage with a Low / Medium / High risk label.
- A ranked list of the top features driving that specific prediction.

This turns the trained models into a usable decision-support tool rather than a static notebook output.

8. Conclusion

This project demonstrates a complete applied machine learning pipeline: from raw data to engineered features, from baseline linear models to non-linear tree-based models, through rigorous evaluation and interpretability analysis, and finally to a deployed interactive tool. The comparison of Logistic Regression and Decision Tree models — and the agreement between their most important features — builds confidence that the identified churn drivers reflect genuine patterns in customer behavior rather than artifacts of a single modeling approach.